

Dawei Wang

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EDUCATION

University of Toronto - Honours Bachelor of Science

Sept 2022 - Jun 2026

- Programs: Immunology Major, Health and Disease Major

SKILLS

Programming: MATLAB, Python, R, FreeSurfer, BIDS, Bash

Laboratory: Mammalian cell culture, transfection, mouse handling, immunohistochemistry, Western blotting, bacterial cell culture, transformation, plasmid isolation, RT-qPCR, cloning (Golden-Gate, Gibson), fluorescence microscopy

RESEARCH EXPERIENCES

Thesis Student - Dept of Neurosciences & Mental Health, Hospital for Sick Children (SickKids)

Sept 2025 - Apr 2026

- Project: Searching for MEG Biomarkers for Concussion
- Supervisor: Dr. Benjamin Dunkley
- Description:
 - Used MATLAB, Python, & graph theory-based modelling to analyze MEG data of resting-state intrinsic neural activity (from seven frequency bands) in adolescent concussion patients and age/sex-matched healthy controls to identify biomarkers for concussion, to improve non-invasive methods for diagnosis of concussion
- Outcomes:
 - Alpha frequency connectivity measures (efficiency, strength, clustering coefficient) was found to be decreased in concussion patients compared to healthy controls
 - No significant changes occurred in beta frequency connectivity measures

Immunology Laboratory Course - University of Toronto

Sept 2025 - Dec 2025

- Course: IMM435 (Practical Immunology)
- Coordinator: Dr. Goetz Ehrhardt
- Description:
 - Practical course teaching experimental procedures commonly used in immunology research; flow cytometry, ELISA, hemolytic plaque assays, hemagglutination assays
 - Conducted weekly lab experiments independently in groups of two, then prepared individual reports due the following week.

Research Assistant - Dept of Neurosciences & Mental Health, Hospital for Sick Children (SickKids)

May 2025 - Aug 2025

- Project: Pilot study investigating the role of glutamate receptors in synaptic plasticity and memory
- Supervisor: Dr. Zhengping Jia
- Description:
 - Investigating the role AMPA receptors and their mutations regulate Alzheimer's development using cell cultures

- Quantified amyloid-beta plaque amounts in transgenic mice using immunohistochemical staining
- Techniques used: Mouse handling and surgery (injection, cardiovascular perfusion), immunohistochemical staining and imaging brain slices, Western blot, HEK293 cell culturing & transfection, fluorescent cell imaging
- Outcomes: No significant difference between AMPAR and APP protein surface expression

Health and Disease Laboratory Course - University of Toronto

Sept 2025 - Dec 2025

- Course: HMB312 (Laboratory in Health and Disease)
- Coordinator: Dr. Alistair Dias
- Description:
 - Laboratory course based on current research techniques and topics which included basic microbiology, molecular biology and animal cell culture techniques, immunocytochemistry, changes in gene expression, and histological techniques
 - Conducted weekly lab experiments independently in groups of two, wrote weekly lab quizzes, and prepared individual reports due two weeks afterward
 - At end of semester, students conducted individual research project based on assigned topic of interest; presented project at course presentation to professor and teaching assistants

POSTERS AND PRESENTATIONS

Child and Youth Mental Health Research Day - Hospital for Sick Children (SickKids)

February 17, 2026

- Conference Speaker
- Title: "Alpha-Frequency Network Disruption as a Biomarker of Adolescent Concussion"

Tri-Campus Undergraduate Research Conference - University of Toronto

February 18, 2026

- Conference Speaker
- Title: "Alpha-Frequency Network Disruption as a Biomarker of Adolescent Concussion"

VOLUNTEER & EXTRACURRICULAR ACTIVITIES

iGEM Research Team Member - University of Toronto, Dept of Biomedical Engineering

Jun 2024 - Nov 2024

- Project: Plasmid.AI
- Supervisor: Dr. Michael Garton (Department of Biomedical Engineering)
- Techniques used:
 - Designed plasmid sequences and primers using Benchling
 - Minipreped *E. coli* sequences and amplified them using PCR, ran sequences on gels between steps to check for desired length
 - Cloned sequences onto vector plasmids using Golden-Gate Assembly
 - Transformed recombinant plasmids into *E. coli* and cultured on antibiotic plate to validate cloning and transformation
- Presented our project at an international competition involving institutions across 50+ countries
 - Placed among the Top 10, the only North American team on the list
 - Project and presentation received a Gold Medal and distinctions of 'Best Model', 'Best Entrepreneurship'

Debate Speaker - iGEM Toronto and Hart House Debate Club

August 27, 2024

- Speaker for debate hosted in collaboration by iGEM research team and the Debate Club to raise awareness on the team's project and research interests
- Researched literature and spoke to experts in AI in medicine, antibiotics, and to gather information in support of my position
- Spoke for affirmative side of debate on the ethics of rapid commercialization of AI in medicine in front of audience of ~25 consisting of students and some visiting faculty

HOSA Competitor - HOSA – Future Health Professionals

May 8, 2024

- Competed
- Received 1st place nationally in the Medical Terminology event at the 9th annual HOSA Spring Leadership Conference, Canada's largest annual health science competition with 6,000+ students participating

Outreach Team Member - iGEM Toronto (Synthetic Biology Research Team)

Mar 2024 - Oct 2024

- In charge of the team's Instagram (@igemtoronto) and email, planned and ran social media campaigns such as 'story takeovers', 'day in the life', and 'collaborative posts' to boost engagement, gaining 600 new followers
- Reached out to external organizations across Canadian universities to collaborate and further raise public awareness on our field of research

Clinical Support Volunteer - Mount Sinai Hospital

Jan 2024 - Dec 2025

- Responsible for holding weekly visits to each of the patients in the geriatric department living with cognitive impairments
- During visits I held activities with the patients; e.g. feeding meals, conversations, playing games, ensuring they were supported and helping maintain their cognitive function

Vice President of Administration - New College Student Council, University of Toronto

Jun 2023 - Apr 2024

- Elected to my university-affiliated college's student council to represent over 6,800 students and be the voice to represent their needs, concerns, and opinions
- Coordinated closely with other executives for student council elections and member transitions, ensuring rules were followed and these events ran as smoothly and efficiently as possible
- Managed and supervised team; supervised executive and member responsibilities and personal conduct, taking appropriate measures against any misconducts

Recognized Study Group Leader - University of Toronto

May 2023 - Jun 2023

- Facilitated weekly organized meetings and activities to review course material
- Prepared weekly studying material, topics of discussion, and activities

Social Chair - New College Residence Council, University of Toronto

Sep 2022 - Apr 2023

- Organized house-wide social events for students in my residence house/floor to increase community engagement among ~50 residents.
- Planned cultural events to better include residents of diverse backgrounds (e.g. Lunar New Year, Diwali).
- Helped plan residence-wide events with the rest of the wide Residence Council to help create a stronger community for the entire New College residence (600+ students)

AWARDS AND DISTINCTIONS

Dean’s List - University of Toronto Apr 2023 - Apr 2026

Awarded to students in good academic standing, with GPA above 3.5

Audrey Taylor Award - University of Toronto New College Registrar Apr 2024

Awarded to students in years 1-3 with good academic standing and significant co-curricular contribution. Valued at \$400

Betty Yeoman Memorial Award - University of Toronto New College Office of Residence Life Jun 2023

Awarded to residents in good academic standing with significant contribution to the residence. Valued at \$1,400